

Congruence & Similarity

Answer sheet · 36 marks total

Section A — Congruence

A1. Congruent: same shape **and** same size (1) similar: same shape but different size / one is an enlargement of the other (1)

A2. (a) **SSS** (1) (b) **RHS** (1) (c) **SAS** (1)

A3. Equal angles only show the triangles are the same shape — one could be bigger than the other, so they are similar but not necessarily congruent (1)

A4. $AB = CD$ and $BC = DA$ (opposite sides of a parallelogram are equal) (1) AC is common to both triangles (1) so congruent by **SSS** (1). Also accept: $AB = CD$, angle $BAC =$ angle DCA (alternate angles), AC common, so **SAS**.

Section B — Similar shapes: missing lengths

B1. $SF = 10 \div 4 = 2.5$ (1) $6 \times 2.5 = \mathbf{15\text{ cm}}$ (1)

B2. $SF = 12.5 \div 5 = 2.5$ (1) $QR = 8 \times 2.5 = \mathbf{20\text{ cm}}$ (1) $PR = 7 \times 2.5 = \mathbf{17.5\text{ cm}}$ (1)

B3. $6 \div 3 = 2$ and $9 \div 5 = 1.8$ (1) ratios are different, so **not similar** (1)

B4. $SF = 12 \div 18 = 2/3$ (1) $24 \times 2/3 = \mathbf{16\text{ cm}}$ (1)

Section C — Similar triangles inside each other

C1. Angle A is shared, and angle $ADE =$ angle ABC (corresponding angles, as DE is parallel to BC), so all three angles are equal (1)

C2. $AB = 4 + 6 = 10$ (1) $SF = 10 \div 4 = 2.5$ (1) $BC = 5 \times 2.5 = \mathbf{12.5\text{ cm}}$ (1)

C3. SF (large to small) $= 6 \div 15 = 0.4$ (1) $DE = 20 \times 0.4 = \mathbf{8\text{ cm}}$ (1)

Section D — Area & volume of similar shapes (Higher)

D1. $3^2 = \mathbf{9}$ (1)

D2. Length $SF = 3$, area $SF = 9$ (1) $10 \times 9 = \mathbf{90\text{ cm}^2}$ (1)

D3. Length $SF = 9 \div 6 = 1.5$ (1) volume $SF = 1.5^3 = 3.375$ (1) $64 \times 3.375 = \mathbf{216\text{ cm}^3}$ (1)

D4. Area $SF = 200 \div 50 = 4$ (1) length $SF = \sqrt{4} = 2$ (1) volume $SF = 2^3 = 8$ (1) $30 \times 8 = \mathbf{240\text{ cm}^3}$ (1)

D5. Cube root of each part: 2 and 3 (1) ratio of heights = $\mathbf{2 : 3}$ (1)

Marking note: award method marks for a correct scale factor even if the final answer contains an arithmetic slip. In proofs, each fact needs a reason to earn its mark; accept any valid congruence condition with correct reasons. Missing units on an otherwise-correct numerical answer should lose the final accuracy mark, not the method mark.